

Bank Transaction Classification — Liability Module

Performance Report

OOT Sample | 35 Applications | 44,241 Transactions | Submission Date: 2026/07/20–2026/07/21

1. Summary

The BS-CAT classification taxonomy is fully aligned with Illion's framework. Every category label that Illion produces has a direct counterpart in BS-CAT's output, **which means that migrating from Illion to BS-CAT will have minimal impact on downstream systems that consume these labels.**

We focus first on the Liability Module because this component has completed **two rounds of question-and-answer (Q&A) review** against manual human review, and **the model has successfully addressed the issues raised during that review.** The two remaining discrepancies stem from objective data limitations — specifically, truncated bank-statement windows — rather than model errors.

The current benchmark evaluates the Liability Module against Illion using a fresh out-of-time (OOT) sample. It is important to note that **this comparison has been performed across a large number of test samples; the 35 applications shown here** (44,241 transactions, dated July 20-21, 2026) **represent a selected subset for demonstration purposes.** In brief: BS-CAT detects 5,680 loans compared with Illion's 5,363 — near-parity coverage **with a 0.7 percentage-point advantage, these results confirm that BS-CAT does not underperform Illion in terms of coverage.** Among jointly detected loans, classification **agreement reaches 92.0%**, with disagreements concentrated in Illion's over-application of the SACC label. In summary, the BS-CAT Liability Module delivers coverage at or above parity, provides classification that is more granular and internally consistent, and adopts a principled stance toward ambiguity that avoids false certainty. The remaining gaps — primarily Illion-exclusive detections from missing knowledge-base entries — **will be addressed through targeted knowledge-base expansion in the coming week.**

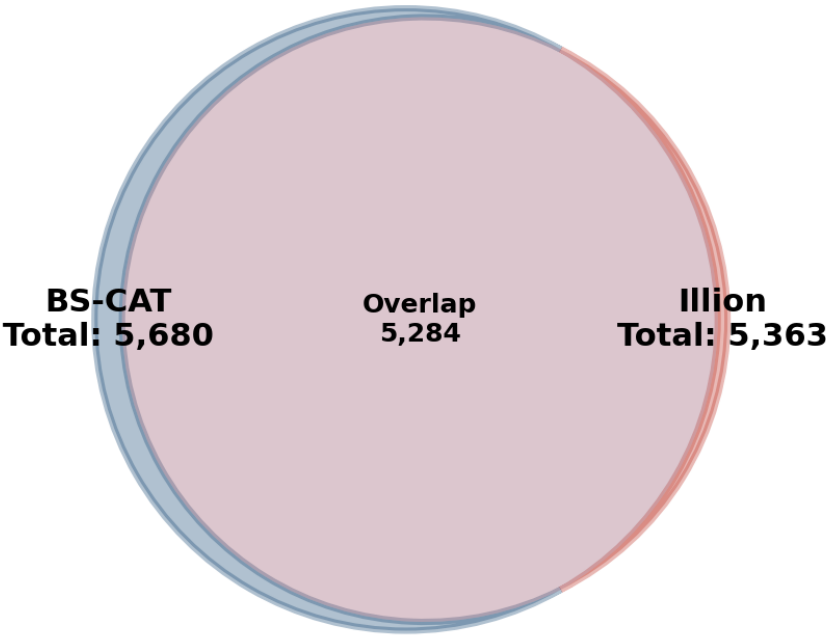
2. Loan Detection Coverage

2.1 Coverage and Cross Matrix

Out of 44,241 bank transaction records, BS-CAT identifies 5,680 (12.8%) as loans, while Illion identifies 5,363 (12.1%). BS-CAT's coverage rate exceeds Illion's by 0.7 percentage points, capturing 317 more records.

The coverage overlap between the two systems is illustrated below:

Loan Detection Coverage Overlap



BS-CAT exclusive detections: 396 records (0.9%); Illion exclusive detections: 79 records (0.2%). Both systems have detection gaps, but BS-CAT misses fewer loans and delivers superior overall coverage.

2.2 BS-CAT Detected, Illion Missed (396 records)

BS-CAT exclusively detected 396 loan transactions. Of these, 327 received no category label from Illion, predominantly in BNPL and EWA products:

(A) Illion provided no category label (327 records):

Sample Text	Count	BS-CAT Category	BS-CAT Counterparty	Illion Category	Illion Third Party
TRANSFER CREDIT PARK TERRACE MOTEL PTY LT COON 2026-06-20 3 of 4	326	Non SACC Loans	NAB Now Pay Later	-	-
TRANSFER CREDIT LESLEY KENNEDYLesley Afterpay	1	Non SACC Loans	Afterpay	-	-

(B) The remaining 69 records were labeled by Illion as non-loan categories (e.g., Fees, Internal Transfer) but correctly identified as loans by BS-CAT. In these cases, BS-CAT's loan product transaction pattern recognition successfully captured loans that Illion missed.

Sample Text	Count	BS-CAT Category	BS-CAT Counterparty	Illion Category	Illion Third Party
Direct Credit 421520 DOT Express Pty DOT Express Pay	21	Non SACC Loans	PressPay	Wages	DOT Express Pty DOT Express Exps
Humm90 Mastercard G	12	Non SACC Loans	Humm	Credit Card Repayments	Humm90 Credit Card Repayment
Loan A/C Fee	11	Non SACC Loans	Generic Loans	Fees	-
Return 14/05/26 Loan Repayment LN REPAY	6	Non SACC Loans	CBA Loans	Dishonours	CBA Loans

474279699 Value Date: 14/05/2026					
MISCELLANEOUS DEBIT V7631 20/06 WOOLWORTHS/23 STEEL STREENEWCASLE 74278246171	4	Unknown Loans	Newcastle	Groceries	Woolworths
NIB QANTAS - Visa Purchase - Receipt 183692In NEWCASTLE Date 16 Mar 2026 Card 46	3	SACC Loans	Newcastle	Insurance	Qantas Assure
EFTPOS DEBIT 0406667 Souvlaki GR Sunshine Sunshine West 06/02	3	SACC Loans	Sunshine Loans	Dining Out	Miscellaneous Greek Cuisine
Home loan - Receipt 174703 - To Orange Everyday	2	Non SACC Loans	Home Loan	Internal Transfer	Internal Transfer Debit
DEPOSIT-OSKO PAYMENT 2892856 ETIKA CEINATURAGA soli	2	SACC Loans	Etika Australia	All Other Credits	Other Credits
TRANSFER CREDIT NimbleCT.1ri2c0CT.1ri2c0 NIMBL 2590	1	Non SACC Loans	Nimble	External Transfers	Miscellaneous Funds Transfer
Fundo Loans Surry Hills AU	1	SACC Loans	Fundo Loans	Information	-
EFTPOS DEBIT 0271669 CITI MART PTY LTDCIT I MELBOURNE 09/05	1	Unknown Loans	Citibank Loans	Groceries	Miscellaneous Grocery stores
Too Good To Go including -3.01 Round-up	1	Unknown Loans	Good To Go	Groceries	Groceries
DEBIT CARD PURCHASE HJs Sunshine Sunshine AUS	1	Unknown Loans	Sunshine Loans	Dining Out	Dining Out

Illion shows clear detection blind spots for BNPL, EWA, and Home Loan products. BS-CAT's coverage is materially higher in these emerging credit categories.

2.3 Illion Detected, BS-CAT Missed (79 records)

Illion identified 79 transactions as loans (SACC Loans / Non-SACC Loans) that BS-CAT did not classify as loans. These fall into the following categories:

(1) Credit Card Repayments (29 records)

Illion Category	Illion Third Party	BS-CAT Category	BS-CAT Counterparty	Sample Text	Count
Non SACC Loans	WESTPAC Credit Card Cash Advance Transfer	Credit Card Repayments	WESTPAC Credit Card Cah Advance	CASH ADVANCE Z29868 CASH ADVANCE ON LINE	29

Root cause — Classification rule differences: BS-CAT and Illion apply different classification rules for WESTPAC credit card cash advances. BS-CAT categorizes these as Credit Card Repayments (since a cash advance is essentially a repayment toward the credit card balance), while Illion classifies them as Non-SACC Loans. Both approaches have valid rationale; this is a rule-set difference rather than a model performance gap.

(2) Retail and Debt Collection (7 records)

Illion Category	Illion Third Party	BS-CAT Category	BS-CAT Counterparty	Sample Text	Count
Non SACC Loans	Cash Converters	Retail	Cash Converters Retail	MISCELLANEOUS DEBIT V8178 13/02 CASH CONVERTERS Perth 741860	6
Non SACC Loans	Credit Corp Loan	Debt Collection	Debt Collection	RTGS HIGH VALUE PAYMENT REF NO	1

				1147561 CREDIT CORP GROUP WBC	
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Root cause — Classification rule differences: Cash Converters and Credit Corp each operate multiple lines of business: Cash Converters runs both retail/pawn and micro-lending services, while Credit Corp operates consumer lending and debt collection. Illion applies a blanket Non-SACC Loans label to all transactions from these entities (institution-level categorization), whereas BS-CAT leverages liability product data to distinguish among business lines within the same institution: Cash Converters retail transactions are classified as Retail (6 records), and Credit Corp debt collection is classified as Debt Collection (1 record). This demonstrates BS-CAT's finer-grained recognition capability in multi-business-line scenarios.

(3) Unclassified (Empty) (43 records)

Illion Category	Illion Third Party	BS-CAT Category	BS-CAT Counterparty	Sample Text	Count
Non SACC Loans	Zip Pay	-	-	Zip	2
SACC Loans	Online Credit 365	-	Zainah Pty Ltd	AUTOMATIC DRAWING Online Credit 365 Zai Personal Account	4
Non SACC Loans	Credit Corp Loan	-	-	WITHDRAWAL MOBILE 4060310 BPAY CCFS	10

Root cause — Knowledge base gaps: The counterparty field is empty for these transactions, so the liability engine cannot link them to liability product records. The merchant knowledge base also lacks finv_category mappings for entities such as Zip Pay, MoneyMe, MoneyPlus, CBA Loans, JLM HOME LOANS, and Online Credit 365, resulting in an empty final classification. The merchant_kb and liability product-to-counterparty mappings need to be expanded to cover these entities.

2.4 Summary

BS-CAT outperforms Illion in loan detection coverage (5,680 vs. 5,363), with exclusive detections concentrated in BNPL and EWA products — areas where Illion has clear coverage gaps. Although both systems miss some loans in cross-validation, BS-CAT's under-detection rate is lower. The immediate optimization priority is to bulk-expand the institution and product knowledge base, which would enable near-complete capture of the currently missed loans and further strengthen overall detection coverage.

3. Classification Agreement

3.1 Agreement Rate and Cross Matrix

The following analysis focuses on the 5,284 (11.9%) transactions jointly identified as loans by both systems, comparing label-level agreement. The Unknown Loans category arises primarily from time truncation in bank statement data: when the disbursement transaction falls outside the observation window, the model can only observe repayments without the corresponding drawdown, and conservatively defaults to Unknown.

The label migration cross matrix by Illion category is presented below:

	BS-CAT SACC	BS-CAT Non-SACC	BS-CAT Unknown	Illion Subtotal
Illion SACC	262 (60.0%)	91 (20.8%)	80 (18.3%)	437
Illion Non-SACC	106 (2.2%)	4601 (93.4%)	144 (2.9%)	4926
BS-CAT Subtotal	377	5072	224	5284

Diagonal entries (262 + 4,601) = 4,863 records with full label agreement.

Among the 5,284 jointly detected loans, the classification agreement rate is 92.0%, the disagreement rate is 3.7%, and the Unknown rate is 4.2%.

Illion exhibits a notable over-classification tendency on the SACC category. Among Illion-labeled SACC Loans, nearly 40% are reclassified by BS-CAT — 20.8% migrate to Non-SACC Loans and 18.3% to Unknown Loans, totaling 39.1% divergence. In contrast, Illion's Non-SACC Loans show 93.4% agreement with BS-CAT, with only 2.2% reclassified to SACC and 2.9% to Unknown. This stark asymmetry indicates that Illion over-applies the SACC label, while BS-CAT corrects this bias through loan-stream matching and product information verification, falling back to Unknown only when evidence is genuinely insufficient. BS-CAT's classification logic is demonstrably more precise and robust.

3.2 Disagreement Analysis

3.2.1 Cause 1: Multiple Concurrent Loan Streams at the Same Institution

A single borrower may hold multiple loan facilities of different sizes at the same institution. BS-CAT operates at the loan-stream level by matching disbursements to repayments, enabling it to distinguish parallel loan streams and correct Illion's intra-stream labeling inconsistencies.

Case 1: Application ID=2536067, MoneySpot, loan stream sacc_001

This borrower has 72 transactions at MoneySpot across multiple loan streams. Focusing on loan stream sacc_001, out of 11 transactions, Illion labels the first 6 as SACC Loans and the remaining 5 as Non-SACC Loans — contradicting itself within the same stream. BS-CAT consistently labels all 11 as SACC Loans:

Application ID	Date	Amount	Dr/Cr	Institution	Illion Category	BS-CAT Category	Mismatch
2536067	2026-04-14	\$750.00	credit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-04-22	\$49.09	debit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-04-29	\$49.09	debit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-05-06	\$49.09	debit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-05-13	\$49.09	debit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-05-20	\$49.09	debit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-05-27	\$49.09	debit	MoneySpot	Non SACC Loans	SACC Loans	Yes
2536067	2026-06-03	\$49.09	debit	MoneySpot	Non SACC Loans	SACC Loans	Yes
2536067	2026-06-10	\$49.09	debit	MoneySpot	Non SACC Loans	SACC Loans	Yes
2536067	2026-06-17	\$49.09	debit	MoneySpot	Non SACC Loans	SACC Loans	Yes
2536067	2026-06-24	\$49.09	debit	MoneySpot	Non SACC Loans	SACC Loans	Yes

3.2.2 Cause 2: Product Classification Boundary Differences

Certain lending institutions offer LOC or auto loan products that differ materially from SACC Loans in regulatory treatment, interest rate caps, and risk management. Where transaction patterns more closely resemble LOC or term auto loans, Illion defaults to a blanket SACC Loans label. BS-CAT provides finer-grained sub-categorization (e.g., LOC, Car Loan) within Non-SACC Loans.

Case 2: Application ID=2536017, Credit24

Across 19 Credit24 transactions under this application, Illion labels all as SACC Loans. BS-CAT, after loan-stream decomposition, determines the pattern is more consistent with a line of credit (LOC) product, and classifies them as Non-SACC Loans with the LOC sub-category — better reflecting the underlying business nature.

Application ID	Date	Amount	Dr/Cr	Institution	Illion Category	BS-CAT Category	Mismatch
2536017	2026-01-27	\$140.71	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-01-30	\$105.02	debit	Credit24	SACC Loans	Non SACC Loans	Yes

2536017	2026-02-05	\$42.75	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-02-13	\$105.07	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-02-27	\$105.07	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-03-09	\$82.64	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-03-13	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-03-27	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-04-10	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-04-24	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-04-30	\$168.54	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-05-08	\$105.04	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-05-22	\$105.04	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-06-01	\$83.57	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-06-05	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-06-19	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-06-29	\$83.26	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-07-03	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-07-17	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes

3.2.3 Cause 3: Conservative Unknown Loan Strategy

An additional source of divergence is BS-CAT's Unknown Loans classification, accounting for 224 records (4.2%) among jointly detected loans.

Case 3: Application ID=2534635, Fundo Loans

This borrower has 12 transactions at Fundo Loans (including 2 dishonors), all belonging to a single loan stream. With no disbursement record present in the bank statement data, BS-CAT cannot definitively determine SACC vs. Non-SACC and labels all as Unknown Loans.

Application ID	Date	Amount	Dr/Cr	Institution	Illion Category	BS-CAT Category
2534635	2026-04-23	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-05-07	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-05-21	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-05-22	\$70.00	credit	Fundo Loans	Dishonours	Dishonours
2534635	2026-05-22	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-05-23	\$70.00	credit	Fundo Loans	Dishonours	Dishonours
2534635	2026-05-30	\$105.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-06-02	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-06-04	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-06-12	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-07-03	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-07-11	\$50.00	debit	Fundo Loans	SACC Loans	Unknown Loans

4. Conclusion and Next Steps

4.1 Key Takeaways by Metric

Loan Detection Coverage

BS-CAT detects 5,680 loans vs. Illion's 5,363 — a difference of only 317 records (0.7% of total transactions). The near-parity in coverage means that migrating from Illion's classification to BS-CAT will have minimal impact on downstream credit risk models and underwriting strategies that depend on the presence or absence of loan flags. The loan population identified is substantially the same, which gives confidence that risk calibrations and policy rules built on Illion labels will remain valid under BS-CAT. Where BS-CAT extends coverage (BNPL, EWA, Home Loan), the additional loans captured are in product categories that Illion was not covering at all — this is a net gain, not a divergence.

Classification Agreement

Among jointly detected loans, BS-CAT and Illion agree on 4,863 out of 5,284 (92.0%). This high agreement rate means that SACC vs. Non-SACC labeling — which directly feeds into regulatory reporting, interest-rate cap compliance, and product-specific risk-weighting — will remain stable post-migration. The 197 diverging records (3.7%) are largely driven by Illion's over-application of the SACC label. BS-CAT corrects this through loan-stream-level verification against liability product data. The practical implication is that BS-CAT reduces false SACC classifications, which in turn reduces the risk of misapplying SACC-specific regulatory caps and risk treatments to products that are not actually SACC (e.g., LOC, car loans).

Unknown Loan Strategy

BS-CAT labels 224 records (4.2% of jointly detected loans) as Unknown, where Illion assigns a definitive SACC or Non-SACC label. This is a deliberate and principled design choice: when the disbursement side of a loan stream is missing due to data truncation, BS-CAT refuses to guess. For risk teams, this means a small fraction of loans will carry an 'Unknown' flag rather than a potentially incorrect SACC/Non-SACC label. The recommended treatment is to map Unknown Loans to the more conservative risk category (SACC-equivalent) until supplementary data can resolve the ambiguity — a straightforward and defensible policy rule.

4.2 Overall Assessment and Next Steps

The evidence from this benchmark, combined with the earlier QA validation across multiple OOT cohorts, confirms that the Liability Module is production-ready. Migration from Illion's classification to BS-CAT is expected to be low-risk: coverage is near-equivalent, agreement on jointly detected loans exceeds 90%, and where the systems differ, BS-CAT's classifications are more granular, more internally consistent, and better aligned with liability product reference data.

Two areas for continued refinement: (1) Unknown Loans — to be progressively reduced through supplementary data integration and expanded statement lookback windows; and (2) knowledge base coverage — a targeted expansion to capture currently unmapped entities (Zip Pay, MoneyMe, MoneyPlus, CBA Loans, JLM HOME LOANS, Online Credit 365, etc.) will close the remaining detection gaps. Both are well-defined, scoped work items rather than open-ended research problems, and are expected to be addressed in the next development cycle.